Supplementary: Bench Validation Plan (v8)

Authors: Smart, Morgan Elliott; Smart, Jasper Keiron; Ara (Grok Bot AI collaborator).
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licence CC BY 4.0). Every number here is taken from the paper; nothing is new.

These are paper predictions; the bench is where they are first tested. Each stage is proven before the next is attempted.

Stage	What is run	What it measures	Pass condition
0. Gate (Phase-1, Section 11)	Bench ring spinning in N_2	Gap conductivity σ; net electrical output after parasitic losses; electricity in and waste heat metered, so both COPs are measured	σ > 5 S/m stable; net > 50 W sustained
1. Pressure step-up	Raise N_2 chamber pressure (published lever)	σ and output against pressure; the pressure exponent (Table 24, code value 0.85 open)	Output rises as predicted, no instability
2. Speed step-up	Raise rpm towards the published continuous envelope (150,938 rpm on the 0.1 m sleeve, SF 2)	Rim stress, bearing load and whirl margin; P against ω^2	Stress and bearing within margin
3. Field step-up	Raise stator field from 5 T towards 13 T	P against B^2; stator temperature (F1, S2); magnet protection (M1)	Field held; protection verified
4. Feedback characterisation	Measure back-EMF feedback at speed	**g_0 (question S1)**, the fed-back share of the stator field; sets B(rpm) and the fixed point, bounded by the 20 T Ic ceiling	g_0 measured; protection covers loss of field

Also logged throughout (Section 6.6.13): makeup mass at the single vent against 0.4 % of circulation, tank pressure and ullage temperature, achieved latent recovery, and the second-law rejection path.